INTERSECTION NUMBERS OF PULLED-BACK INVERTIBLE SHEAVES SCALE BY THE DEGREE OF THE MORPHISM RESTRICTED TO THE SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites nine cross-chapter labels (including a definition), each recorded in assembly.external_references with its canonical tag as a navigation pointer.

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Allowed standard prerequisites (editorial).

- Numerical polynomial leading term and the Euler characteristic along a morphism. Stacks Project, Varieties, Lemmas lemma-numerical-polynomial-leading-term (tag 0BEN) and lemma-euler-characteristic-morphism (tag 0BEK). The first reduces the comparison to leading coefficients and the second gives the alternating-sum expansion of the Euler characteristic along f.
- Dimension theory for locally algebraic schemes and generic finiteness. Stacks Project, Varieties, Lemmas lemma-dimension-locally-algebraic (tag 0A21) and lemma-dimension-fibres-locally-algebraic (tag 0B2L); Morphisms, Lemma morphisms-lemma-generically-finite (tag 02NW). Shows the generic point of Z is the unique point over the generic point of f(Z) and that f|_Z is generically finite, which makes the degree available.
- Vanishing of higher direct images of a finite morphism and the projection formula. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-finite-pushforward-coherent (tag 01Y6); Cohomology, Lemma cohomology-lemma-projection-formula (tag 01E8). Gives the vanishing of R^i f_*O_Z near xi and the rewriting of the twisted Euler characteristic.

Source reference. Original reference 02NY: sources/stacks-chow/tags.txt, source lines 3497–3497. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BEK: sources/stacks-chow/tags.txt, source lines 14353–14353. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01E8: sources/stacks-chow/tags.txt, source lines 1825–1825. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BEM: sources/stacks-chow/tags.txt, source lines 14355–14355. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0A21: sources/stacks-chow/tags.txt, source lines 12689–12689. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B2L: sources/stacks-chow/tags.txt, source lines 13934–13934. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02NW: sources/stacks-chow/tags.txt, source lines 3495–3495. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01Y6: sources/stacks-chow/tags.txt, source lines 2545–2545. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BEN: sources/stacks-chow/tags.txt, source lines 14356–14356. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let $f : Y \to X$ be a morphism of proper schemes over k. Let $Z \subset Y$ be an integral closed subscheme of dimension d and let $\mathcal{L}_1, \ldots, \mathcal{L}_d$ be invertible $\mathcal{O}_X$-modules. Then*

$$(f^*\mathcal{L}_1 \cdots f^*\mathcal{L}_d \cdot Z) = \deg(f|_Z : Z \to f(Z))(\mathcal{L}_1 \cdots \mathcal{L}_d \cdot f(Z))$$

where $\deg(Z \to f(Z))$ *is as in Morphisms, Definition 02NY or* 0 *if* $\dim(f(Z)) < d$.

Proof. The left hand side is computed using the coefficient of $n_1 \ldots n_d$ in the function

$$\chi(Y, \mathcal{O}_Z \otimes f^*\mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes f^*\mathcal{L}_d^{\otimes n_d}) = \sum (-1)^i \chi(X, R^i f_* \mathcal{O}_Z \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_d^{\otimes n_d})$$

The equality follows from Lemma 0BEK and the projection formula (Cohomology, Lemma 01E8). If $f(Z)$ has dimension $< d$, then the right hand side is a polynomial of total degree $< d$ by Lemma 0BEM and the result is true. Assume $\dim(f(Z)) = d$. Let $\xi \in f(Z)$ be the generic point. By dimension theory (see Lemmas 0A21 and 0B2L) the generic point of Z is the unique point of Z mapping to ξ. Then $f : Z \to f(Z)$ is finite over a nonempty open of $f(Z)$, see Morphisms, Lemma 02NW. Thus $\deg(f : Z \to f(Z))$ is defined and in fact it is equal to the length of the stalk of $f_*\mathcal{O}_Z$ at ξ over $\mathcal{O}_{X,\xi}$. Moreover, the stalk of $R^i f_* \mathcal{O}_X$ at ξ is zero for $i > 0$ because we just saw that $f|_Z$ is finite in a neighbourhood of ξ (so that Cohomology of Schemes, Lemma 01Y6 gives the vanishing). Thus the terms $\chi(X, R^i f_* \mathcal{O}_Z \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_d^{\otimes n_d})$ with $i > 0$ have total degree $< d$ and

$$\chi(X, f_* \mathcal{O}_Z \otimes \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_d^{\otimes n_d}) = \deg(f : Z \to f(Z))\chi(f(Z), \mathcal{L}_1^{\otimes n_1} \otimes \ldots \otimes \mathcal{L}_d^{\otimes n_d}|_{f(Z)})$$

modulo a polynomial of total degree $< d$ by Lemma 0BEN. The desired result follows. □